Ahmad Hossein Yazdani

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Summary

My research focuses on designing efficient **Systems for ML (particularly LLMs)** and applying **ML/LLMs to optimize system performance**. I am also interested in emerging areas such as **adapting distributed applications to persistent memory environments**, **GPU scheduling for distributed workloads**, and **software–hardware co-design** to enhance serverless computing.

Work Experience

August,2020 – present **_Research Assistant at Distributed System and Storage Lab_, Virginia Tech**.

Advisor : Dr. Ali Butt, *Professor, Department of Computer Science*, Virginia Tech

◦ **Metis:** The project introduces a data access pattern for Deep Learning workloads to improve the cachability of the deep learning workloads. With a small cache, the project improves the cache hit ratio by up to 4.5×compared to the LRU caching policy.

◦ **User I/O profiling:** Led a collaborative research with Analytics & AI Methods at Scale Group at Oak Ridge National Laboratory (ORNL) on analytically recognizing the behavior of the users and jobs submitted to HPC systems to improve the I/O efficiency of the HPC systems. The analysis revealed that the I/O pattern of a users' job submissions **can be predicted over a window of 10 days with an accuracy of above 90%** based on the application-level I/O statistics and the applications' I/O patterns

◦ **LLMStore:** A collaborative research project with Analytics & AI Methods at Scale Group at Oak Ridge National Laboratory (ORNL) and Lawrence Berkeley National Laboratory **for LLM fine-tuning aiming to minimize offloading parameters, optimizer state and activation offloading in HPC using caching and deduplication solutions**. This project is carried out in collaboration with Jean Luca Bez, Ahmad Maroof Karimi, Arnab Kumar Paul, Suren Byna. Our initial analysis revealed **a traffic of over 300 GB for tensor offloading** to the storage for LLama finetuning.

◦ **Silo-based open framework for Federated Learning (2025-):** Contributing to a **Federated Learning research** project led by Dr. Feiyi Wang (ORNL), Dr. Ali Butt (Virginia Tech), and Dr. Sahil Tyagi (ORNL). The project integrates Federated Learning across cloud and HPC environments, enabling geographically distributed clients to collaborate with privacy-preserving, hierarchical gradient aggregation and low communication overhead. My role focuses on **implementing efficient collective gradient aggregation and compression** within the gRPC communication layer to reduce cross-institution communication costs.

January,2026 – ***Research Intern, Oak Ridge National Laboratory***.

Mentors: William Godoy, Pedro Vallero

◦ Throughout this internship, I am planning to investigate the use of AI tools (Like ChatHPC) towards optimization of HPC workflows with huge memory and storage pressure.

June,2024 – August,2024 ***Student Assistant at NERSC, Lawrence Berkeley National Laboratory (LBNL), internship***.

Mentors: Stephen Simms, Lisa Gerhardt, Jean Luca Bez

○ I investigated the causes of I/O hotspots in HPC applications and analyzed common performance issues. Specifically, I examined Drishti, an HPC I/O recommendation tool, and found it generates many false positive warnings, and derived insights to improve the accuracy of the the Drishti I/O recommendation tool.

June,2023 – August,2023 ***Student Assistant at Lawrence Berkeley National Laboratory (LBNL), internship***.

Mentors: Suren Byna, Jean Luca Bez

○ I continued my research on characterizing the sources of I/O performance variation in HPC, and striving to alleviate the I/O performance variability. I found out a significant I/O variability for various HPC applications like E3SM (For earth modeling) and LAMMP (For molecular simulations). Later continued my research for Large AI models and language models which I'm working on in the LLMStore project

June,2021 – August,2021 ***Internship at Oak Ridge National Laboratory, Analytics & AI Methods at Scale Group***.

Mentors: Feiyi Wang, Sarp Oral, Ahmad Maroof Karimi and Arnab Kamur Paul

○ I studied and characterized the application I/O pattern using clustering techniques, and then extracted features from their submitter like the job runs of the same application over a time-window for that user, the scale of the job submissions the submitter tends to submit. **This work resulted in my ICDCN'25 paper, where I was able to predict the I/O pattern of the next job given the features from the past submissions of the same user with an accuracy of nearly 90% for HPC jobs.**

Computer skills

Programming Languages & frameworks: Python, R, C, C++, Advanced JAVA EE, JAVA SE, Tensorflow, Go, Rust, MPI, pthread, OpenMP, CUDA, Java bytecode, X86 assembly

Machine Learning & GPU programming: PyTorch, keras, scikit-learn, DeepSpeed, LLM, Computer Vision, OmniFed

Systems: Linux kernel programming, distributed systems, File Systems, Lustre, GPFS, Slurm, MapReduce, Redis, Ceph, NVMe, Darshan, Docker, Kubernetes, Large Language Models, Deep Learning, Federated Learning

Conference & Workshop publications

[ICDCN'25] **Yazdani, Ahmad Hossein**, Arnab K. Paul, Ahmad Maroof Karimi, Feiyi Wang, and Ali Butt. User-based i/o profiling for leadership scale hpc workloads. In *Proceedings of the 26th International Conference on Distributed Computing and Networking*, ICDCN '25, page 181–190, New York, NY, USA, Jan. 2025. Association for Computing Machinery. doi:10.1145/3700838.3700865.

[**FAST'23★**] Redwan Ibne Seraj Khan, **Yazdani, Ahmad Hossein**, Yuqi Fu, Arnab K Paul, Bo Ji, Xun Jian, Yue Cheng, and Ali R Butt. Shade: Enable fundamental cacheability for distributed deep learning training. In *Proceedings of the 21th USENIX Conference on File and Storage Technologies*, page 14, Santa Clara, CA, US, Feb. 2023. USENIX Association. URL `https://www.usenix.org/conference/fast23/presentation/khan`.

★ Top-tier venue

Education

2020–present **PhD, Computer Science**, *Virginia Polytechnic Institute and State University (Virginia Tech)*, Blacksburg, VA, US.
Advisor: Dr Ali Butt

May 2025 **Masters of Computer Science**, *Virginia Polytechnic Institute and State University (Virginia Tech)*, Blacksburg, VA, US.
Advisor: Dr Ali Butt

2015–2020 : **Bachelor of Computer Software Engineering**, *University of Tehran*, Tehran, Iran.

Notable teaching experience

Virginia Tech

Fall 2023, 2024 **CS3214: Computer Systems, head TA**.

- Served as the head TA; creating the rubrics for the assignments and coordinating the logistics.

Fall 2022, spring 2023 **CS3214: Computer Systems, instructor**.

- Giving presentations to one section (75 students) in parallel with two other sections taught by Godmar Back and Huaicheng Li.